Assessment Tool 2

Short answer Test

1) For the grammar

$E \rightarrow E+T \mid T$ remove left recursion

Given:

$E \rightarrow E+T \mid T$

The general transformation is:

$A \rightarrow A\alpha \mid B$

becomes

$A \rightarrow \beta A'$

$A' + \alpha A' \mid \varepsilon$

Therefore:

$E \rightarrow TE'$

$\boxed{E' \rightarrow +TE' \mid \varepsilon}$

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2) Find FIRST(E) for:

$E \rightarrow TE', E' \rightarrow +TE' \mid \varepsilon$

Given:

$E \rightarrow TE'$

$E' \rightarrow TE' \mid \varepsilon$

Since E begins with T:

FIRST(E) = FIRST(T)

If T represent an identifier or number. for example:

$T \rightarrow id$

then:

$\boxed{\text{FIRST}(E) = \{id\}}$

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3) For the grammar $E \rightarrow T+E \mid T$

check if it is ambiguous or not

Given:

$E \rightarrow T+E \mid T$

This grammar is not ambiguous for string consisting of T' separated by +

for example:

T+T+T

has only the right-associative derivation:

E => T+E => T+T+E => T+T+T

Therefore

The grammar is unambiguous.

4> Identify whether string aba is valid for:

S->aSa|b

Given

S->aSa|b

Derivation:

S => aSa

=> aba

Therefore:

$\boxed{\text{'aba' is valid}}$

5> Check whether the string ab is generated by:

S->aA, A->b

Given:

S->aA

A->b

Derivation:

S => aA

=> ab

Therefore:

'ab' is generated

6> Construct a parse tree for a+b*c using:

E->E+E|E*E|id

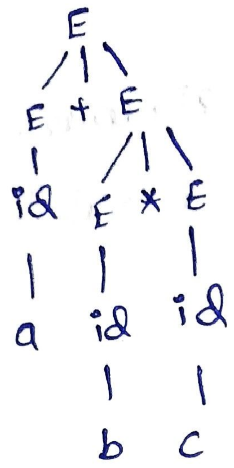
Grammar:

E->E+E|E*E|id

The grammar is ambiguous, but using the conventional precedence where * has higher precedence than +, the intended structure is:

a+(b*c)

Parse Tree:



Thus $\boxed{a+(b*c)}$

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7) Apply left factoring for:

$S \to aA \mid aB \mid b$

Given

$S \to aA \mid aB \mid b$

The common prefix is a

Therefore

$S \to aS' \mid b$

$\boxed{S' \to A \mid B}$

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8) Find FOLLOW(A) for:

$S \to AB, B \to b$

Given

$S \to AB$

$B \to b$

Since A is followed by B:

FOLLOW(A) = FIRST(B)

Since:

FIRST(B) = {b}

Therefore:

$\boxed{\text{FOLLOW}(A) = \{b\}}$

9) Given the grammar
$S \to aSb \mid \varepsilon$

Given:
$S \to aSb \mid \varepsilon$

Derivation:

$$
\begin{aligned}
S &\Rightarrow aSb \\
&\Rightarrow aaSbb \\
&\Rightarrow aaaSbbb \\
&\Rightarrow aaa\varepsilon bbb \\
&\Rightarrow aaabbb
\end{aligned}
$$

Therefore:

'aaabbb' is valid.

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10) Convert the following ambiguous grammar into an unambiguous grammar:
$E \to E+E \mid E*E \mid id$

Given:
$E \to E+E \mid E*E \mid id$

Assume:
* has higher precedence than +
Both operators are left associative

An unambiguous grammar is

$$
\begin{aligned}
E &\to E+T \mid T \\
T &\to T*F \mid F \\
F &\to id
\end{aligned}
$$

This ensures that multiplication has higher precedence than addition.